

Summary of main points from last lecture

$$J_{ij} = \frac{\lambda}{N} s_i^0 s_j^0 + E_{ij}$$

$$E_{ij} \sim \mathcal{N}(0, 1/N)$$

$$s_i^0 = \pm 1 \text{ with equal prob}$$

$$E(\vec{s}) = - \sum_{1 \leq i < j \leq N} J_{ij} s_i s_j$$

$$P_\beta(\vec{s}) = \frac{e^{-\beta E(\vec{s})}}{Z(\beta, \lambda, E, s^0)}$$

$$-\beta F(\beta, \lambda, E, s^0) = \ln Z(\beta, \lambda, E, s^0)$$

$$-\beta F(\beta, \lambda) = \langle \ln Z(\beta, \lambda, E, s^0) \rangle_{E, s^0}$$

$$\langle \ln Z \rangle = \lim_{n \rightarrow 0} \frac{\langle Z^n \rangle - 1}{n} \quad (Z^n = e^{n \ln Z} \xrightarrow{n \rightarrow 0} 1 + n \ln Z)$$

$$\langle Z^n \rangle_{E, s^0} = \sum_{\vec{s}^0} \dots \sum_{\vec{s}^n} \left\langle e^{\beta \sum_{i,j} \sum_{a=1}^n \left[\frac{\lambda}{N} s_i^a s_j^a + E_{ij} \right] s_i^a s_j^a} \right\rangle_E$$

$$\langle Z^n \rangle_{E, s^0} = \sum_{\vec{s}^0} \dots \sum_{\vec{s}^n} e^{-NE(Q'_{ab})}$$

After averaging over E , replicated partition function involves a sum over replicated spins where the quantity being summed depends on spins only through overlaps Q'_{ab}

$Q'_{ab} = n+1$ by $n+1$ matrix of overlaps:

$$Q'_{ab} = \begin{bmatrix} 1 & m_1 & m_2 & \dots & m_n \\ m_1 & 1 & & & \\ m_2 & & 1 & & \\ \vdots & & & \ddots & \\ m_n & & & & 1 \end{bmatrix}$$

$0 \leq a, b \leq n$

$$Q'_{ab} = \begin{cases} m_b & a=0, 1 \leq b \leq n \\ Q_{ab} & 1 \leq a \neq b \leq n \\ 1 & a=b \end{cases}$$

$$m_a \equiv \frac{1}{N} \sum_{i=1}^N s_i^0 s_i^a \quad 1 \leq a \leq n$$

$$Q_{ab} \equiv \frac{1}{N} \sum_{i=1}^N s_i^a s_i^b \quad 1 \leq a, b \leq n$$

$$E[Q'] = - \left(\beta \frac{\lambda}{2} \sum_{a=1}^n m_a^2 + \frac{\beta n}{4} + \frac{\beta^2}{2} \sum_{1 \leq a < b \leq n} Q_{ab} \right)$$

signal term coming
from s^0 pulling
all s^a towards it

noise term coming
from E_{ij} pulling
all s^a close to
each other

$$\langle z^n \rangle = e^{N \{-E[Q'] + S[Q']\}}$$

where

$$e^{N S[Q']} = \sum_{\vec{s}^0} \dots \sum_{\vec{s}^n} \prod_{0 \leq a < b \leq n} \delta \left[\sum_{i=1}^N s_i^a s_i^b - N Q'_{ab} \right]$$

$S[Q']$ = entropy of configurations $\vec{s}^0, \dots, \vec{s}^n$ with given overlap Q'

$$e^{N S[Q']} = \int \prod_{0 \leq a < b \leq n} \pi d\hat{Q}'_{ab} e^{N \left\{ - \sum_{0 \leq a < b \leq n} \hat{Q}'_{ab} Q'_{ab} + \Phi[\hat{Q}'] \right\}}$$

Saddle point approx to integral over $\hat{Q}'_{ab} \Rightarrow S[Q']$ is Legendre transform of $\Phi[\hat{Q}']$

$$\Phi[\hat{Q}'] = \ln Z[\hat{Q}']$$

$$Z[\hat{Q}'] = \sum_{s^0, \dots, s^n} e^{\sum_{0 \leq a < b \leq n} \hat{Q}'_{ab} s^a s^b}$$

(negative) free
energy

Partition
function

of n spin
replica system

$$\langle z^n \rangle = \int \prod_{0 \leq a < b \leq n} \pi d\hat{Q}'_{ab} dQ'_{ab} e^{-N \mathcal{F}[\hat{Q}', Q']}$$

$$\mathcal{F}[\hat{Q}', Q'] = E[Q'] + \sum_{0 \leq a < b \leq n} \hat{Q}'_{ab} Q'_{ab} - \Phi[\hat{Q}']$$

now

$$-\beta F(\beta, \lambda) = \langle \ln Z \rangle = \lim_{n \rightarrow 0} \frac{\langle z^n \rangle - 1}{n}$$

if $\langle z^n \rangle = e^{f(n)}$ with $f(0) = 0$

$$\lim_{n \rightarrow 0} \frac{\langle z^n \rangle - 1}{n} = \frac{1 + \frac{\partial f}{\partial n} \Big|_{n=0} n - 1}{n}$$

$$\lim_{n \rightarrow 0} \frac{\langle z^n \rangle - 1}{n} = \frac{\partial f}{\partial n} \Big|_{n=0}$$

we perform $N \rightarrow \infty$ limit by taking saddle point approx to integrals over $\hat{Q}'_{ab}$ and Q_{ab}
then take $n \rightarrow 0$ limit by keeping terms linear in n in exponent

$$\frac{\beta F(\beta, \tau)}{N} = \lim_{n \rightarrow 0} \text{extr}_{\hat{Q}', Q'} \mathcal{F}[\hat{Q}', Q']$$

here by $n \rightarrow 0$ we mean
keep terms linear in n
and take the coefficient of
the linear term

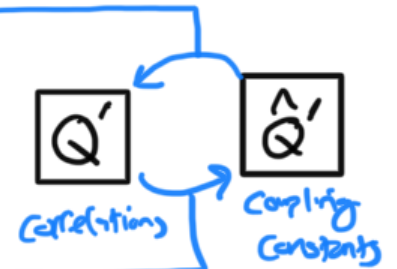
• Saddle point equations

$$\frac{\partial}{\partial \hat{Q}'_{ab}} \mathcal{F} = 0 \Rightarrow$$

$$Q'_{ab} = \langle s^a s^b \rangle_{\hat{Q}'}$$

$$\frac{\partial}{\partial Q_{ab}} \mathcal{F} = 0 \Rightarrow$$

$$\hat{Q}'_{ab} = \frac{\partial E}{\partial Q'_{ab}}$$



$\langle \cdot \rangle_{\hat{Q}'}$ is an avg w.r.t. distribution

$$P(s^0, s^1, \dots, s^n) = \frac{1}{Z[\hat{Q}']} e^{+\sum_{0 \leq a < b \leq n} \hat{Q}_{ab} s^a s^b}$$

Physical meaning of order parameters

$\vec{s}^1, \vec{s}^2$ drawn iid from $P_3(\vec{s})$

$$q = \frac{1}{N} \sum_{i=1}^N s_i^1 s_i^2$$

$$m \equiv \frac{1}{N} \sum_{i=1}^N s_i^0 s_i^1$$

$$P(q) = \lim_{n \rightarrow 0} \frac{1}{n(n-1)} \sum_{1 \leq a < b \leq n} \delta[q - Q_{ab}]$$

$$P(m) = \lim_{n \rightarrow 0} \frac{1}{n} \sum_{a=1}^n \delta[m - m_a]$$

After averaging out disorder E

End of review

Now lets solve saddle point equations, take $n \rightarrow 0$ and compute free energy

$$Q'_{ab} \equiv \begin{bmatrix} 1 & m_1 & m_2 & \dots & m_n \\ m_1 & 1 & & & \\ m_2 & & 1 & & \\ \vdots & & & \ddots & \\ m_n & & & & 1 \end{bmatrix}$$

$$\hat{Q}'_{ab} \equiv \begin{bmatrix} 1 & \hat{m}_1 & \hat{m}_2 & \dots & \hat{m}_n \\ \hat{m}_1 & 1 & & & \\ \hat{m}_2 & & 1 & & \\ \vdots & & & \ddots & \\ \hat{m}_n & & & & 1 \end{bmatrix}$$

$$\mathcal{F}[\hat{Q}', Q'] = E[Q'] + \sum_{0 \leq a < b \leq n} \hat{Q}'_{ab} Q'_{ab} - \Phi[\hat{Q}']$$

lets make a replica symmetric ansatz:

$$\begin{aligned} m_a &= m & \hat{m}_a &= m \\ Q_{ab} &= q & \hat{Q}_{ab} &= \hat{q} \end{aligned}$$

$$E[Q'] = -\beta \sum_{a=1}^n m_a^2 = -\frac{\beta^2 n}{2} = -\frac{\beta^2}{2} \sum_{a=1}^n Q_{ab}^2$$

$$= -\beta \frac{\lambda}{2} n m^2 - \frac{\beta^2 n}{4} - \frac{\beta^2}{2} \underbrace{\frac{n(n-1)}{2}} q$$

$$\frac{n^2}{2} - \frac{n}{2} \rightarrow -\frac{n}{2} \text{ as } n \rightarrow 0$$

$$\rightarrow n \left[-\frac{\beta^2}{4} - \beta \frac{\lambda}{2} m^2 + \frac{\beta^2}{4} q^2 \right]$$

↑ note: minimizing $E(m, q)$ w.r.t q

makes q larger for $n > 1$

but makes q smaller for $n < 1$

$$\sum_{0 \leq a < b \leq n} \hat{Q}'_{ab} Q'_{ab} = \sum_{a=1}^n \hat{m}_a m_a + \sum_{1 \leq a, b \leq n} \hat{Q}_{ab} Q_{ab}$$

$$= n \hat{m} m + \frac{n(n-1)}{2} \hat{q} q$$

$$\rightarrow n \left[\hat{m} m - \frac{1}{2} \hat{q} q \right]$$

Now for phi

$\Phi[\hat{Q}'_{ab}]$ is a bit more involved:

$$\Phi[\hat{Q}'] = \ln \sum_{s^0 \dots s^n} e^{\sum_{0 \leq a < b \leq n} \hat{Q}'_{ab} s^a s^b}$$

$$\sum_{0 \leq a < b \leq n} \hat{Q}'_{ab} s^a s^b = \sum_{b=1}^n \hat{Q}'_{0b} s^0 s^b + \sum_{1 \leq a < b \leq n} \hat{Q}'_{ab} s^a s^b$$

$$= \frac{1}{M} \sum_{b=1}^n s^0 s^b + \hat{q} \sum_{1 \leq a < b \leq n} s^a s^b$$

$$\frac{1}{2} \left[\sum_a s^a \right]^2 - \frac{n}{2}$$

$$= \frac{1}{M} \sum_{b=1}^n s^0 s^b + \frac{\hat{q}}{2} \left[\sum_a s^a \right]^2 - \frac{n}{2} \hat{q}$$

$$\Phi[\hat{M}, \hat{q}] = \ln \sum_{s^0 \dots s^n} e^{\hat{M} s^0 \sum_{b=1}^n s^b + \frac{\hat{q}}{2} \left[\sum_a s^a \right]^2 - \frac{n}{2} \hat{q}}$$

couple the replicas

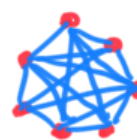
decouple them via Hubbard-Stratonovich

transform:

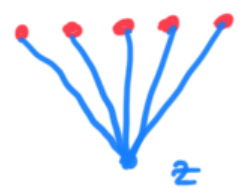
$$e^{\frac{\hat{q}}{2} \sum_{a,b} s^a s^b} = \int \mathcal{D}z \sqrt{\hat{q}} e^{z \sum_a s^a}$$

$$\mathcal{D}z \equiv \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

Gaussian measure



All to all coupling of strength $\hat{q}$



feed forward Gaussian input w/

$$\Phi[\hat{M}, \hat{q}] = \ln e^{-\frac{n}{2} \hat{q}} \int \mathcal{D}z \sum_{s^0 \dots s^n} e^{(\hat{M} s^0 + \sqrt{\hat{q}} z) \sum_{a=1}^n s^a}$$

replicas decoupled !!

$$\Phi[\hat{M}, \hat{q}] = -\frac{n}{2} \hat{q} + \ln \int \mathcal{D}z \sum_{s^0} \left[\sum_s e^{(\hat{M} s^0 + \sqrt{\hat{q}} z) s} \right]^n$$

$Z(h) =$ partition function of a single spin in a field $h = \hat{M} s^0 + \sqrt{\hat{q}} z$

(keep coefficient of term linear in n in Taylor expansion in $n=0$)

now take $n \rightarrow 0$

$z = \text{gaussian variable}$
 $Z = \text{partition function}$

note: if $f(n) = \ln \int \mathcal{D}z Z(z)^n$

$$= \ln \int \mathcal{D}z e^{n \ln Z(z)}$$

$$\approx \ln \int \mathcal{D}z \underbrace{\left[1 + n \ln Z(z) \right]}_{=0}$$

$$n \int \mathcal{D}z \ln Z(z)$$

so $\ln \int \mathcal{D}z Z(z)^n \xrightarrow{n \rightarrow 0} n \int \mathcal{D}z \ln Z(z)$

so as $n \rightarrow 0$

$$\Phi[\hat{m}, \hat{q}] = n \left\{ -\frac{\hat{q}}{2} + \int \mathcal{D}z \sum_{s_0} f[\hat{m} s_0 + \sqrt{\hat{q}} z] \right\}$$

$$f[h] = \ln \sum_s e^{h s}$$

$$= \ln 2 \cosh(h)$$

$f[h] = (\text{negate})$ free energy of a single spin in a magnetic field of strength h (at $KT=1$)

Putting it all together

$$\frac{\beta F(\beta, \lambda)}{N} = \text{extr}_{\hat{q}, \hat{m}, \hat{m}} \left(-\frac{\beta^2}{4} - \frac{\beta \lambda}{2} m^2 + \frac{\beta^2}{4} q^2 \right) +$$

← from $E[Q']$

$$\left(\hat{m} m - \frac{1}{2} \hat{q} q \right) +$$

← from $\sum_{a < b} Q'_{ab} Q'_{ab}$

$$\left(+\frac{\hat{q}}{2} - \int \mathcal{D}z \sum_{s_0} f[\hat{m} s_0 + \sqrt{\hat{q}} z] \right)$$

← from $-\Phi[Q']$

$$\frac{\partial}{\partial m} : \hat{m} = \beta \lambda m$$

$$\frac{\partial}{\partial q} : \frac{\beta^2}{2} q = \frac{\hat{q}}{2}$$

$$\hat{q} = \beta^2 q$$

need to compute derivatives of this with respect to $\hat{m}$ and $\hat{q}$:

lets solve a more general problem once and for all:

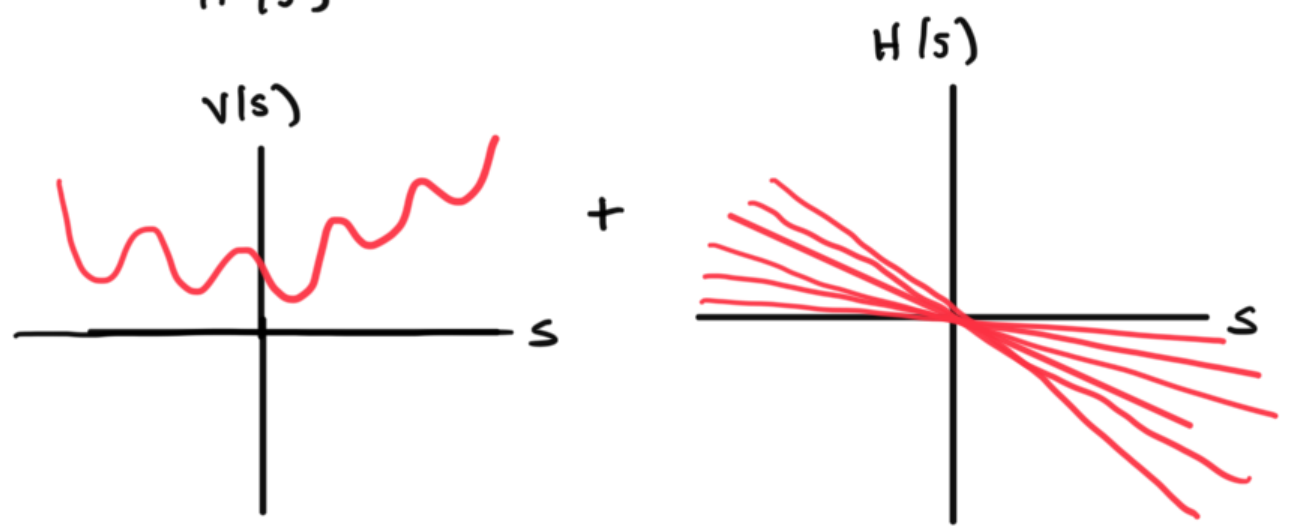
• Let s be a general scalar variable (ie $s \in \mathbb{R}$ or s is discrete)

• Let s be subject to an orbiting potential $v(s)$ such

that $p(s) \propto e^{-\beta v(s)}$

- Let's tilt the energy landscape by turning on a random electric field with additional energy

$$H(s) = -\hat{\mu}s - \sqrt{\hat{q}}zs \quad z \sim \mathcal{N}(0,1)$$



$$\vec{E} = -\frac{\partial H}{\partial s} = \hat{\mu} + \sqrt{\hat{q}}z$$

$\hat{\mu}$ = mean value of electric field

$\sqrt{\hat{q}}$ = standard deviation of electric field

now let

$$f(\hat{\mu}, \hat{q}) \equiv \int \mathcal{D}z \ln \int ds e^{-\beta[V(s) - \hat{\mu}s - \sqrt{\hat{q}}zs]}$$

free energy averaged over all values of the electric field

$f(\hat{\mu}, \hat{q} | z) = -1/\beta \times$ free energy of system for a particular value of the electric field given by z

above we have a special case: for a fixed s_0 :

$$\int \mathcal{D}z \ln \sum_s e^{(\hat{\mu}s_0 + \sqrt{\hat{q}}z)s}$$

here $s \rightarrow s = \pm 1$

$$\int ds \rightarrow \sum_s$$

$$V(s) \rightarrow 0$$

$$\hat{\mu} \rightarrow \hat{\mu}s_0$$

$$\hat{q} \rightarrow \hat{q}$$

$$s \rightarrow 1$$

now we want to understand derivatives of $f(\hat{\mu}, \hat{q})$

General case special case

$$\begin{aligned} \frac{\partial f}{\partial \hat{\mu}} &= \frac{\partial}{\partial \hat{\mu}} \int \mathcal{D}z \ln \int ds e^{-\beta[V(s) - \hat{\mu}s - \sqrt{\hat{q}}zs]} \\ &= \beta \int \mathcal{D}z \frac{\int ds s e^{-\beta[V(s) - \hat{\mu}s - \sqrt{\hat{q}}zs]}}{Z(\beta, \hat{\mu}, \hat{q}, z)} \end{aligned}$$

$$Z(\beta, \hat{\mu}, \hat{q}, z) \equiv \int ds e^{-\beta[V(s) - \hat{\mu}s - \sqrt{\hat{q}}zs]}$$

$$\frac{\partial f}{\partial \hat{\mu}} = \beta \int \mathcal{D}z \langle s | z \rangle_{\hat{\mu}, \hat{q}}$$

where $\langle \cdot | z \rangle_{\hat{\mu}, \hat{q}}$ is an average w.r.t the distribution

$$p(s|z) = \frac{e^{-\beta [v(s) - \hat{\mu}s - \sqrt{\hat{q}}zs]}}{Z[\beta, \hat{\mu}, \hat{q}, z]}$$

$$\begin{aligned} \frac{\partial f}{\partial \hat{q}} &= \frac{\partial}{\partial \hat{q}} \int \mathcal{D}z \ln \int ds e^{-\beta [v(s) - \hat{\mu}s - \sqrt{\hat{q}}zs]} \\ &= \int \mathcal{D}z \int ds + \frac{\frac{1}{2} \hat{q}^{-1/2} \beta z s e^{-\beta [v(s) - \hat{\mu}s - \sqrt{\hat{q}}zs]}}{Z[\beta, \hat{\mu}, \hat{q}, z]} \end{aligned}$$

$$= \frac{1}{2} \hat{q}^{-1/2} \beta \int \mathcal{D}z z \int ds s \frac{e^{-\beta [\dots]}}{Z}$$

$$= \frac{1}{2} \hat{q}^{-1/2} \beta \int \mathcal{D}z \int ds s \partial_z \left[\frac{e^{-\beta [\dots]}}{Z} \right]$$

$$= \frac{1}{2} \hat{q}^{-1/2} \beta \int \mathcal{D}z \int ds s \left[\frac{z \beta \hat{q}^{1/2} s e^{-\beta [\dots]} - e^{-\beta [\dots]} \frac{\partial z}{\partial z}}{z^2} \right]$$

Now use Stein's lemma

$$\langle z F(z) \rangle = \langle \frac{\partial F}{\partial z} \rangle$$

$$z \sim \mathcal{N}(0, 1)$$

(recall)

$$\partial_z \left[\frac{F}{g} \right] = \frac{g \partial_z F - F \partial_z g}{g^2}$$

$$= \frac{1}{2} \beta^2 \int \mathcal{D}z \left\{ \frac{\int ds s^2 e^{-\beta [\dots]}}{Z} - \left[\frac{\int ds s e^{-\beta [\dots]}}{Z} \right]^2 \right\}$$

$$\begin{aligned} \frac{\partial z}{\partial z} &= \frac{\partial}{\partial z} \int ds e^{-\beta [\dots]} \\ &= \int ds \beta \hat{q}^{1/2} s e^{-\beta [\dots]} \end{aligned}$$

$$= \frac{1}{2} \beta^2 \int \mathcal{D}z \left[\langle s^2 | z \rangle_{\hat{\mu}, \hat{q}} - \langle s | z \rangle_{\hat{\mu}, \hat{q}}^2 \right]$$

$$= \frac{1}{2} \beta^2 \int \mathcal{D}z \left[\text{Var}(s | z)_{\hat{\mu}, \hat{q}} \right]$$

Summary

$$f(\hat{\mu} + \sqrt{\hat{q}}z) = \ln \int ds e^{-\beta [v(s) - (\hat{\mu} + \sqrt{\hat{q}}z)s]}$$

$-\beta$ times free energy of a particle s in an arbitrary potential v plus an electric field $\hat{\mu} + \sqrt{\hat{q}}z$

$$f(\hat{m}, \hat{q}) = \int \mathcal{D}z + (\hat{m} + \sqrt{\hat{q}}z) \rightarrow \text{average free energy over random electric field } (z \sim \mathcal{N}(0,1))$$

$$\frac{\partial f}{\partial \hat{m}} = \beta \int \mathcal{D}z \langle s | z \rangle_{\hat{m}, \hat{q}} \rightarrow \text{a usual cumulant relation: derivative of free energy w.r.t. } \hat{m} \text{ (coefficient of } s \text{ in energy) is the first cumulant of } s \text{ (its mean) but now averaged over random } z$$

$$\frac{\partial f}{\partial \hat{q}} = \frac{\beta^2}{2} \int \mathcal{D}z \text{Var}(s | z)_{\hat{m}, \hat{q}}$$

a generalized cumulant relation: derivative of free energy w.r.t. variance $\hat{q}$ of the electric field is average variance of s

$$\langle \cdot | z \rangle_{\hat{m}, \hat{q}} \text{ is on average w.r.t. distribution}$$

$$p(s | z, \hat{m}, \hat{q}) \equiv \frac{e^{-\beta [v(s) - (\hat{m} + \sqrt{\hat{q}}z)s]}}{Z[\beta, \hat{m}, \hat{q}, z]}$$

Now back to our specific problem

$$\frac{\beta F(\beta, \gamma)}{N} = \text{extr}_{\hat{q}, \hat{q}, m, \hat{m}} \left[-\frac{\beta^2}{4} - \frac{\beta\gamma}{2} m^2 + \frac{\beta^2}{4} q^2 + \hat{m}m - \frac{1}{2} \hat{q}(1-q) - \int \mathcal{D}z \sum_{s_0} p(s_0) f[\hat{m}s_0 + \sqrt{\hat{q}}z] \right]$$

Saddle point equations

$$\frac{\partial}{\partial m} : \hat{m} = \beta\gamma m$$

$$\frac{\partial}{\partial \hat{m}} : m = \int \mathcal{D}z \sum_{s_0} p(s_0) s_0 \langle s | \hat{m}s_0 + \sqrt{\hat{q}}z \rangle$$

$$\frac{\partial}{\partial q} : \hat{q} = \beta^2 q$$

$$\frac{\partial}{\partial \hat{q}} : \frac{1-q}{2} = \frac{1}{2} \int \mathcal{D}z \sum_{s_0} p(s_0) \text{Var}(s | \hat{m}s_0 + \sqrt{\hat{q}}z)$$

recall a single spin s in an external field h

$$P(s|h) = \frac{e^{hs}}{Z[h]} = \frac{e^{hs}}{e^{hs} + e^{-hs}} = \frac{e^{hs}}{2 \cosh h}$$

$$\langle s | h \rangle = \sum_s s P(s|h) = \frac{e^h - e^{-h}}{2 \cosh h} = \tanh(h)$$

$$\text{Var}(s|h) = \langle s^2 | h \rangle - \langle s | h \rangle^2 = 1 - [\tanh(h)]^2$$

then the saddle point equations simplify to:

$\hat{m} = \beta \lambda m$	$m = \int \mathcal{D}z \sum_{s_0} p(s_0) s_0 \langle s \hat{m} s_0 + \sqrt{q} z \rangle$
$\hat{q} = \beta^2 q$	$q = \int \mathcal{D}z \sum_{s_0} p(s_0) \langle s \hat{m} s_0 + \sqrt{q} z \rangle^2$

where $\langle s | h \rangle = \tanh(h)$

We can eliminate $\hat{m}$ and $\hat{q}$

$m = \int \mathcal{D}z \sum_{s_0} p(s_0) s_0 \tanh(\beta [\lambda m s_0 + \sqrt{q} z])$
$q = \int \mathcal{D}z \sum_{s_0} p(s_0) \tanh^2(\beta [\lambda m s_0 + \sqrt{q} z])$

• need to solve these equations for q and m as a function of β and λ

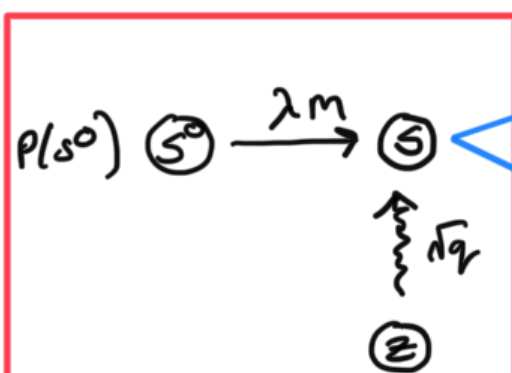
can also eliminate $\hat{m}$ and $\hat{q}$ from free energy:

$$\frac{\beta F(\beta, \lambda)}{N} = \text{extr}_{q, \hat{q}, m, \hat{m}} \left[-\frac{\beta^2}{4} - \frac{\beta \lambda}{2} m^2 + \frac{\beta^2}{4} q^2 + \underbrace{\hat{m} m}_{\beta \lambda m^2} - \underbrace{\frac{1}{2} \hat{q} (q-1)}_{-\frac{1}{2} \beta^2 q (q-1)} - \int \mathcal{D}z \sum_{s_0} p(s_0) f[\hat{m} s_0 + \sqrt{\hat{q}} z] \right]$$

$\frac{\beta F(\beta, \lambda)}{N} = \text{extr}_{q, m} \left[-\frac{\beta^2}{4} + \frac{\beta \lambda}{2} m^2 - \frac{\beta^2}{4} q^2 + \frac{1}{2} \beta^2 q - \int \mathcal{D}z \sum_{s_0} p(s_0) f(\beta [\lambda m s_0 + \sqrt{q} z]) \right]$	furth simplification: note $f(h) = f(-h)$ $\mathcal{D}z \rightarrow \mathcal{D}z$ under $z \rightarrow -z$
$-\frac{\beta^2}{4} (q^2 - 2q)$ $-\frac{\beta^2}{4} [(q-1)^2 - 1]$ $-\frac{\beta^2}{4} (q-1)^2 + \frac{\beta^2}{4}$	$\Rightarrow \int \mathcal{D}z \sum_{s_0} p(s_0) f(\beta [\lambda m s_0 + \sqrt{q} z])$ $= \int \mathcal{D}z f(\beta [\lambda m + \sqrt{q} z])$

$\frac{\beta F(\beta, \lambda)}{N} = \text{extr}_{q, m} \left[\frac{\beta \lambda}{2} m^2 - \frac{\beta^2}{4} (q-1)^2 - \int \mathcal{D}z f(\beta [\lambda m + \sqrt{q} z]) \right]$

Picture behind mean field theory



$$m = \langle \langle s | s_0, z \rangle s_0 \rangle_{z, s_0}$$

$$q = \langle \langle s | s_0, z \rangle^2 \rangle_{z, s_0}$$

two self consistent conditions

$$N(0,1)$$

A single spin s driven by:

- 1) a random field z with coupling $\sqrt{q}$
- 2) a random field s^0 with coupling λm

or even simpler:

$$\begin{aligned} \textcircled{s} \\ \uparrow \\ h \sim N(\lambda m, q) \end{aligned} \quad \begin{aligned} m &= \langle \langle s|h \rangle \rangle_h \\ q &= \langle [\langle s|h \rangle]^2 \rangle_h \end{aligned}$$

To understand phase diagram, we need to find solutions $m^*(\beta, \lambda)$ and $q^*(\beta, \lambda)$

$$\begin{aligned} m &= \int Dz \tanh \beta [\lambda m + \sqrt{q} z] \\ q &= \int Dz \tanh^2 \beta [\lambda m + \sqrt{q} z] \end{aligned}$$

warm up: say $\lambda=0$ (pure spin glass) $J_{ij} \sim N(0, 1/N)$
first equation then $\Rightarrow m=0$

$$q = \int Dz \tanh^2 \beta \sqrt{q} z$$

$\tanh x \approx x$ for small x

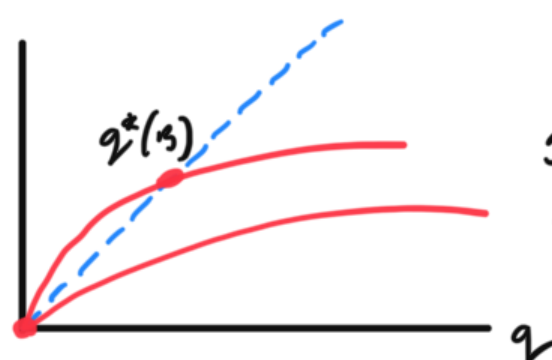
$q=0$ is always a soln

high temperature limit $\beta \rightarrow 0$ $q \approx \int Dz \beta^2 q z^2 = \beta^2 q$
or small q

for $\beta < 1$ $KT > T$ $q=0$ is the only solution

--- unity line

— $\int Dz \tanh^2 \beta \sqrt{q} z$



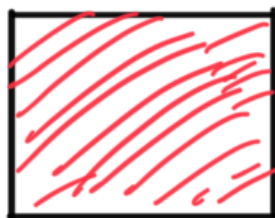
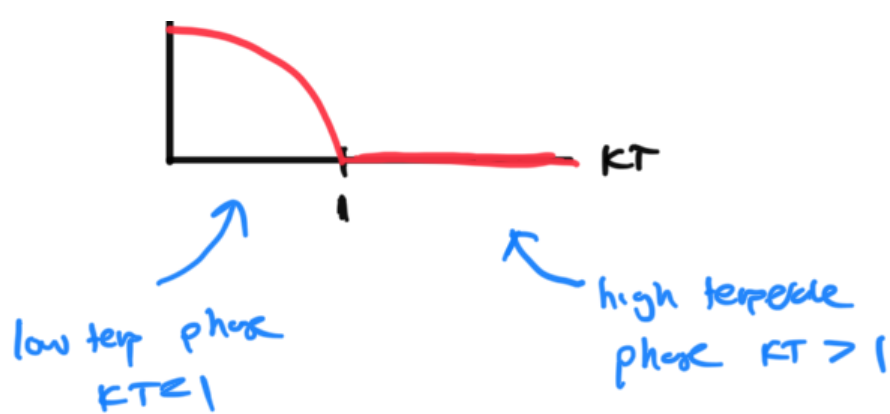
$\beta > 1$ $KT < T$ low temp
 $\beta < 1$ $KT > T$ high temp

$$q^*(KT)$$

Recall

N_{-1-2}

out of 2 real solutions



$$q \equiv \frac{1}{N} \sum_{i=1}^N S_i S_i$$

$$P(q) = \delta[q - q^*(\beta)]$$

replica symmetric prediction for distribution of q

$$q^* = \frac{1}{N} \sum_{i=1}^N \langle S_i^2 \rangle$$

$P_\beta(\vec{S})$ confined

$$P[q] = \delta[q - q^*(\beta)]$$

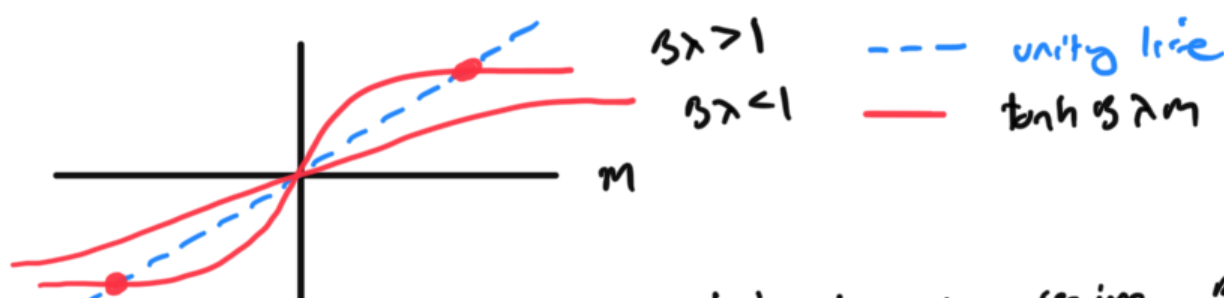
$P_\beta(\vec{S})$ spread over all of spin config space

$$P[q] = \delta[q]$$

Now let $\lambda \neq 0$

- note $q=0$ and $m=0$ are always solutions
- let $q=0$ and search for other solutions in m :

$m = \tanh(\beta \lambda m)$ we recover mean field equation for a ferromagnet with coupling λ

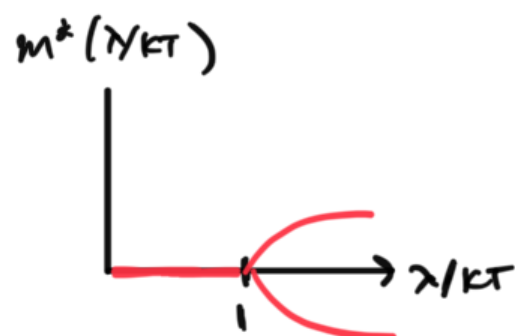


high temp regime $\beta \lambda < 1$ or $KT > \lambda$

$$m^* = 0$$

low temp regime $\beta \lambda > 1$ or $KT < \lambda$

$$m^* \neq 0$$



(all this is assuming $q=0$)

recall $m \equiv \frac{1}{N} \sum_{i=1}^N S_i^0 S_i$

$$P[m] = \delta[m - m^*]$$

To find solutions when q and $m \neq 0$ we must use numerics



$P_\beta(\vec{S})$ spread out

$$P[q] = \delta[q]$$

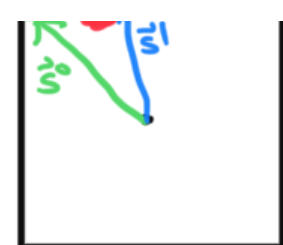
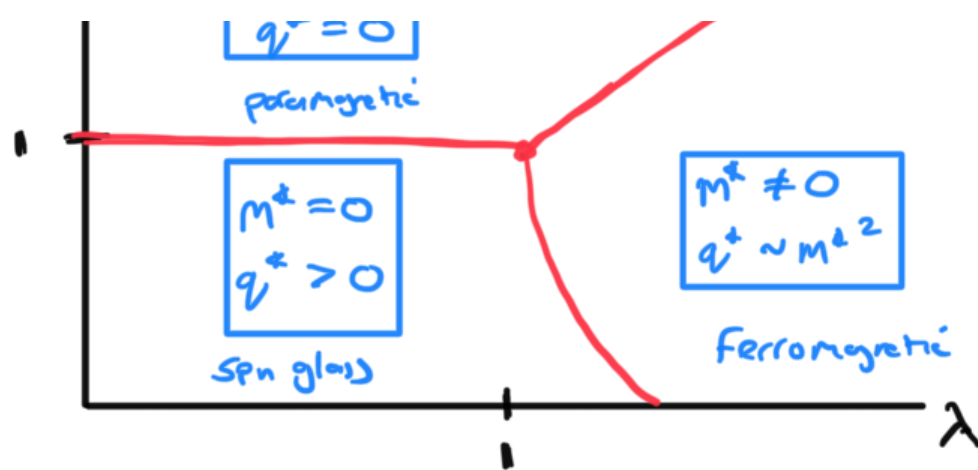
$$P[m] = \delta[m]$$

KT

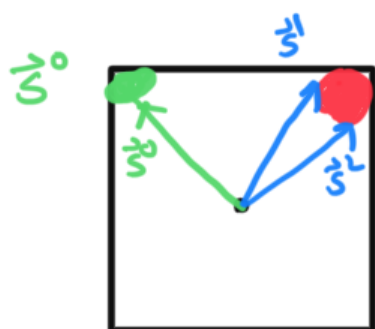
$$m^* = 0$$

$\vec{s}^0$





$P_3(\vec{s})$ is correlated to $\vec{s}^0$ but is not more correlated than that:
 $q^* = m^{*2}$
 ie $\langle s_i \rangle = m$
 $\Rightarrow q^* = \frac{1}{N} \sum_{i,j} \langle s_i \rangle \langle s_j \rangle = m^2$



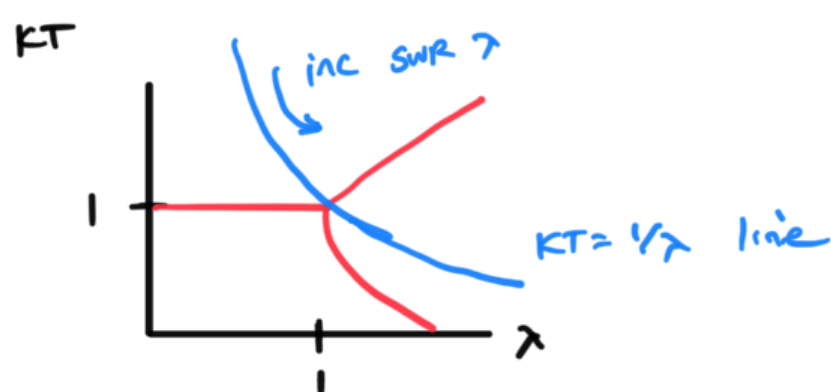
$P_3(\vec{s})$ is correlated: $P[q] = \delta[q - q^*(\lambda, \tau)]$

but is not correlated to s^0 : $P[m] = \delta[m]$

Back to Bayesian inference:

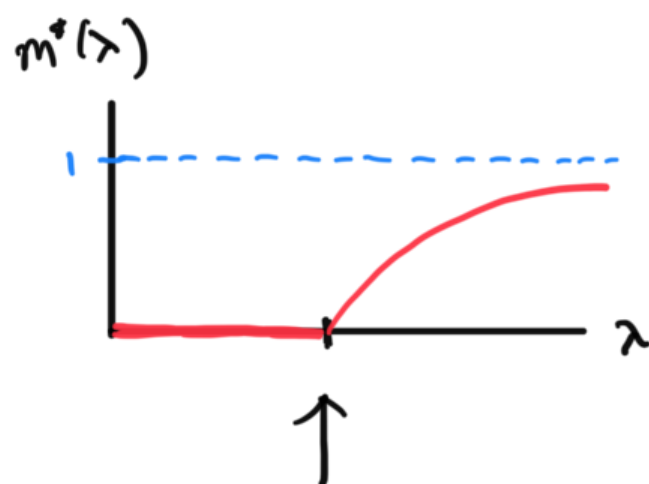
$\hat{s}_\beta \equiv \int d\vec{s} \vec{s} P_3(\vec{s})$ this is the posterior mean iff $\beta = \lambda$

or $\boxed{KT = \frac{1}{\lambda}}$



performance of Bayesian estimator $\hat{s}_\beta = \lambda$ as a function of SWR λ

can be obtained from $m^*(\lambda)$



a phase transition in performance of a Bayesian algorithm !!

for $\lambda < 1$ $\vec{s}^0$ is buried in the connectivity
 $J_{ij} = \lambda s_i^0 s_j^0 + E_{ij}$

Bayesian inf cannot recover it

for $\lambda > 1$ $\vec{s}^0$ can start to be recovered...

